Tertiary Infotech Academy Pte Ltd

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LEARNER GUIDE

For

Microsoft Certified Endpoint Administrator Associate (MD-102)

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Conducted by

Tertiary Infotech Academy Pte Ltd

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
2.0	15 July 2024	Prior accredited MD-102 courseware release.	Dr Alfred Ang
3.0	25 August 2026	Full technical rebuild aligned to the MD-102 skills measured 24 July 2026; added five-domain coverage, ten connected labs, Graph/PowerShell automation, Security Copilot, Intune Suite and evidence-led troubleshooting.	Dr Alfred Ang
3.1	25 August 2026	Expanded the five-day trainer deck beyond 550 slides with control-contract and diagnostic-trace deep dives for every objective; regenerated the slide map and Lesson Plan references.	Dr Alfred Ang

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1. Course purpose and alignment

This Learner Guide supports Microsoft Certified Endpoint Administrator Associate (MD-102) (TGS-2024042603), a 5-day, 40-hour WSQ course mapped to Security Administration (ICT-OUS-3012-1.1). The technical scope follows the Microsoft MD-102 skills measured as of 24 July 2026 while preserving the accredited learning outcomes and K/A criteria.

Component	Hours	Evidence
Classroom facilitation	19	Mechanism, configuration and diagnostic interpretation
Practical	19	Ten connected labs with evidence packs
Assessment	2	60-min WA + 60-min PP

1.1 Learning outcomes

- LO1: Administer complex security programs and analyze system-update impacts on Microsoft Endpoint administration.
- LO2: Perform Microsoft Endpoint system administration and configure authentication to protect against security threats.
- LO3: Troubleshoot Microsoft Endpoint security issues and assist users in defining access rights.
- LO4: Coordinate advanced access-control rights and investigate unauthorized-access incidents on Microsoft Endpoint systems.

1.2 Current exam blueprint

Domain	Weight	WSQ alignment
Prepare infrastructure for devices	20–25%	LO1 · K1 · K4 · A1 · A3
Manage and maintain devices	25–30%	LO1 · LO2 · K1 · K2 · A2 · A4
Protect devices	15–20%	LO2 · LO3 · K3 · K4 · A4 · A5 · A6
Manage and secure applications	15–20%	LO2 · LO3 · K2 · K3 · A2 · A5
Optimize endpoint operations	10–15%	LO3 · LO4 · K3 · K5 · A5 · A7 · A8

1.3 Evidence standard

- Configured state: exact policy, assignment, field, command or script.
- Observed result: service and client state, including timestamp or correlation ID.
- Diagnosis: first failing stage supported by a log, report, error or absent state.
- Recovery: smallest safe change followed by a reproducible acceptance check.

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2. Prepare infrastructure for devices (20–25%)

Alignment: LO1 · K1 · K4 · A1 · A3. The following sections mirror the deck's mechanism → contract → failure/control → verification pattern.

2.1 Choose the correct Microsoft Entra device identity

Mechanism

1. User sign-in: Primary refresh token
2. Registration service: Device object
3. Join state: Registered · joined · hybrid
4. Token claims: deviceid · trustType
5. Access decision: Identity plus device state

Exact contract or configuration

Field/state	Required interpretation
Workplace join	AzureAdRegistered = YES
Cloud join	AzureAdJoined = YES
Hybrid join	DomainJoined + AzureAdJoined
Verification	dsregcmd /status

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Pending device	Check registration event log
Wrong tenant	Inspect TenantId and DeviceId
Missing PRT	Validate WAM and sign-in
Duplicate object	Reconcile stale registrations

Verification evidence

- Device State block: Join flags
- SSO State block: AzureAdPrt = YES
- Entra device record: Owner and trust type
- Sign-in log: Device ID correlation
- Practice bridge: Lab 1.

2.2 Target policy with dynamic device groups

Mechanism

1. Device inventory: Extension attributes
2. Rule parser: Property operators

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3. Membership engine: Asynchronous evaluation

4. Assignment resolver: Include and exclude

5. Policy service: Per-device applicability

Exact contract or configuration

Field/state	Required interpretation
OS rule	device.deviceOSType -eq "Windows"
Ownership rule	device.deviceOwnership -eq "Company"
Autopilot rule	device.devicePhysicalIDs -any _ -contains "[ZTDId]"
Preview	Validate rules before save

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Invalid property	Use supported device schema
Slow membership	Check processing status
Nested group assumption	Dynamic groups do not use nested evaluation
Broad targeting	Test with a preview cohort

Verification evidence

- Rule syntax: Saved expression
- Preview result: Expected devices
- Membership audit: Added/removed objects
- Assignment status: Policy reached cohort
- Practice bridge: Lab 1.

2.3 Configure Windows automatic enrollment

Mechanism

1. Licensed user: Intune entitlement

2. MDM user scope: Enrollment authority

3. Entra join: Discovery URL

4. Enrollment client: Certificate request

5. Intune record: Managed device object

Exact contract or configuration

Field/state	Required interpretation
MDM scope	Some or All
Discovery URL	EnterpriseEnrollment.manage.microsoft.com
Enrollment status	dsregcmd /status
Diagnostics	MDMDiagReport.html

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Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Scope excludes user	Review MDM user scope
License missing	Assign Intune entitlement
Stale enrollment	Inspect scheduled tasks
Network block	Test enrollment endpoints

Verification evidence

- ManagedBy value: Intune
- Enrollment certificate: MS-Organization-Access
- Event log: DeviceManagement-Enterprise-Diagnostics-Provider
- Intune record: Last check-in
- Practice bridge: Lab 2.

2.4 Control multi-platform enrollment

Mechanism

1. Platform authority: Apple · Google · Samsung
2. Enrollment token: Profile or QR
3. Ownership signal: Corporate or personal
4. Restriction evaluation: Platform + OS limits
5. Managed record: Compliance-capable device

Exact contract or configuration

Field/state	Required interpretation
Apple corporate	ABM automated enrollment
Android corporate	Fully managed or corporate work profile
Android shared	Dedicated device
Restriction	Block personal where required

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Expired token	Renew before service impact
Wrong ownership	Use corporate identifiers
Unsupported OS	Check minimum version
Profile conflict	Verify one intended enrollment path

Verification evidence

- Token status: Active and unexpired
- Enrollment profile: Assigned devices

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- Ownership field: Corporate
- Platform report: Enrollment success
- Practice bridge: Lab 2.

2.5 Delegate Intune administration safely

Mechanism

1. Role definition: Permissions
2. Role assignment: Members
3. Scope group: Objects administered
4. Scope tag: Objects visible
5. Audit log: Accountability

Exact contract or configuration

Field/state	Required interpretation
Built-in role	Least-privilege baseline
Custom role	Only required actions
Scope tag	Regional visibility boundary
Multi-admin approval	Second-person control

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Role too broad	Compare actions to task
Empty scope group	Assignment has no reach
Tag mismatch	Object not visible
Approval deadlock	Maintain approver coverage

Verification evidence

- Effective permissions: Expected actions only
- Visibility test: Only tagged objects
- Approval record: Requester and approver
- Audit event: Role change trace
- Practice bridge: Lab 1.

2.6 Evaluate compliance before access

Mechanism

1. Device reports state: Intune check-in
2. Compliance engine: Policy evaluation
3. Entra claim: isCompliant

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4. Conditional Access: Grant control

5. Resource token: Allow or block

Exact contract or configuration

Field/state	Required interpretation
Grace period	Controlled remediation window
Mark noncompliant	Immediate or delayed
CA grant	Require compliant device
Break-glass	Exclude emergency account

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Not evaluated	Check assignment and check-in
False block	Inspect per-setting state
CA lockout	Use report-only first
Stale signal	Force sync and reassess

Verification evidence

- Per-setting status: Failed condition
- Device compliance: Timestamp
- CA result: Grant control detail
- Sign-in correlation: User + device + policy
- Practice bridge: Lab 3.

2.7 Deploy Windows Hello for Business and Windows LAPS

Mechanism

1. Identity bootstrap: MFA proof

2. Key provisioning: TPM-backed credential

3. Policy delivery: Cloud Kerberos trust

4. Local admin rotation: LAPS policy

5. Recovery access: Audited retrieval

Exact contract or configuration

Field/state	Required interpretation
WHfB trust	Cloud Kerberos trust
PIN policy	Length and anti-spoofing
LAPS backup	Microsoft Entra ID
Rotation	After-use password reset

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Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Provisioning blocked	Check tenant and device prerequisites
TPM unavailable	Review attestation and policy
Password not backed up	Inspect LAPS event log
Overprivileged retrieval	Restrict role and audit

Verification evidence

- WHfB container: certutil -DeleteHelloContainer for reset
- dsregcmd: NgcSet = YES
- LAPS status: Password backup timestamp
- Audit log: Password retrieval actor
- Practice bridge: Lab 3.

2.8 Troubleshoot enrollment as an evidence chain

Mechanism

1. User eligibility: License and scope
2. Device eligibility: Platform and ownership
3. Discovery: MDM endpoints
4. Client execution: Enrollment task
5. Service record: Managed device state

Exact contract or configuration

Field/state	Required interpretation
Local report	MDMDiagReport.html
Event channel	Admin + Operational logs
Service view	Troubleshooting + support
Network test	Required endpoint reachability

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
0x80180014	Personal Windows blocked
0x80180018	User not authorized
0x801c03f3	Device registration problem
Duplicate serial	Retire stale record

Verification evidence

- Error code: Exact HRESULT
- Correlation ID: Service-side trace

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- Enrollment log: First failing stage
- Remediation retest: New successful record
- Practice bridge: Lab 2.

3. Manage and maintain devices (25–30%)

Alignment: LO1 · LO2 · K1 · K2 · A2 · A4. The following sections mirror the deck's mechanism → contract → failure/control → verification pattern.

3.1 Select Windows Autopilot or device preparation

Mechanism

1. Corporate identity: Hardware hash or preparation policy
2. Profile assignment: Device or user group
3. OOBE contact: Cloud discovery
4. Enrollment: Entra + Intune
5. Provisioning: Apps and policies

Exact contract or configuration

Field/state	Required interpretation
Classic Autopilot	Registered corporate devices
Device preparation	Streamlined cloud-native setup
User-driven	Named-user productivity
Self-deploying	Shared or kiosk scenario

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
No profile	Verify assignment state
Hash mismatch	Re-import correct identity
Unsupported mode	Check TPM and network
Policy race	Use readiness reporting

Verification evidence

- Deployment profile: Assigned
- Device identity: Registered
- Enrollment report: Phase timings
- Final state: Entra joined and managed
- Practice bridge: Lab 4.

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3.2 Control Autopilot with ESP and naming

Mechanism

1. OOBE: Tenant identification
2. Device setup: Security policies
3. Account setup: User-targeted payload
4. ESP gate: Blocking selection
5. Desktop: Provisioning complete

Exact contract or configuration

Field/state	Required interpretation
Name template	Prefix plus serial or random
Block use	Until required apps install
Timeout	Controlled failure route
Tracking	Autopilot deployment report

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Infinite ESP	Review blocking apps
Invalid name	Respect 15-character limit
App dependency	Validate install order
User reaches desktop early	Correct ESP targeting

Verification evidence

- ESP phase: Device or account
- Blocking app: Install status
- Enrollment time: Phase duration
- Device name: Template resolved
- Practice bridge: Lab 4.

3.3 Upgrade Windows 11 with deployment rings

Mechanism

1. Inventory: Readiness signals
2. Pilot ring: Early adopters
3. Validation: App and driver health
4. Broad rings: Progressive rollout
5. Rollback window: Recovery control

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Exact contract or configuration

Field/state	Required interpretation
Feature policy	Target Windows release
Safeguard hold	Respect by default
Deadline	User experience boundary
Rollback	Uninstall period

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Compatibility hold	Inspect safeguard ID
Low disk	Use readiness report
Driver failure	Pause affected cohort
Productivity impact	Roll back within window

Verification evidence

- Readiness report: Capable or blocked
- Feature version: Target achieved
- Failure code: Deployment error
- Ring KPI: Success and rollback rate
- Practice bridge: Lab 7.

3.4 Provision Windows 365 Cloud PCs

Mechanism

1. License: Cloud PC entitlement
2. Provisioning policy: Image + network
3. Group assignment: User scope
4. Provisioning service: Cloud PC creation
5. Management: Intune lifecycle

Exact contract or configuration

Field/state	Required interpretation
Join type	Microsoft Entra joined
Network	Microsoft-hosted or ANC
Image	Gallery or custom
Resize/reprovision	Controlled lifecycle action

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Provisioning failed	Review network and license

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Image validation	Fix generalized image
Connectivity issue	Check ANC health
Data loss risk	Understand reprovision impact

Verification evidence

- Provisioning state: Succeeded
- Connectivity check: Healthy
- Intune record: Cloud PC managed
- End-user test: Windows 365 connection
- Practice bridge: Lab 4.

3.5 Translate GPOs into Settings Catalog policy

Mechanism

1. GPO export: XML report
2. Group Policy analytics: Import
3. CSP mapping: Supported settings
4. Policy creation: Settings Catalog
5. Assignment: Cloud-managed cohort

Exact contract or configuration

Field/state	Required interpretation
Migration readiness	Supported · deprecated · unknown
ADMX import	Custom administrative templates
OMA-URI	Only when catalog lacks setting
Conflict state	Multiple sources disagree

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Unsupported setting	Redesign control
Duplicate config	Remove overlapping profile
ADMX dependency	Import parent first
Conflict	Trace per-setting source

Verification evidence

- Analytics report: Migration percentage
- Policy JSON: Exact settings
- Per-setting status: Success/conflict/error
- Client MDM report: Applied CSP
- Practice bridge: Lab 5.

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3.6 Target profiles with filters and enrollment-time grouping

Mechanism

1. Base assignment: User or device group
2. Filter query: Device properties
3. Include/exclude: Assignment logic
4. Enrollment-time group: Immediate classification
5. Policy delivery: Resolved applicability

Exact contract or configuration

Field/state	Required interpretation
Filter mode	Include or exclude
Property	manufacturer · model · ownership
Evaluation	At assignment check
Enrollment category	Provisioning-time cohort

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Unsupported property	Use filter schema
Wrong operator	Test sample values
User assignment mismatch	Know device filter context
Race condition	Use enrollment-time grouping

Verification evidence

- Filter preview: Expected match
- Assignment status: Filter applied
- Device property: Exact value
- Policy result: Correct cohort only
- Practice bridge: Lab 5.

3.7 Apply Intune Suite controls

Mechanism

1. Privilege request: User asks elevation
2. EPM policy: Rule evaluation
3. Audit signal: Elevation telemetry
4. Support channel: Remote Help
5. Certificate/network: Cloud PKI or Tunnel

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Exact contract or configuration

Field/state	Required interpretation
EPM rule	File hash · publisher · path
Remote Help	RBAC and session consent
Cloud PKI	Issuing CA + SCEP profile
Tunnel for MAM	Per-app protected path

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Elevation denied	Rule mismatch
Remote Help unavailable	License or RBAC
Certificate failure	Chain or connector health
Tunnel offline	Gateway health and routing

Verification evidence

- Elevation report: File + user + rule
- Session audit: Helper and sharer
- Certificate profile: Issued identity
- Tunnel report: Connection and gateway
- Practice bridge: Lab 5.

3.8 Perform remote actions and live device query

Mechanism

1. Operator action: RBAC authorization
2. Intune service: Command queue
3. Device channel: Push or check-in
4. Client execution: Action result
5. Audit/report: Outcome recorded

Exact contract or configuration

Field/state	Required interpretation
Retire	Remove company data
Wipe	Reset device
Fresh Start	Reinstall Windows
Device query	KQL against live device data

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Action pending	Device offline

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Wrong destructive action	Confirm ownership and intent
Query unavailable	Check supported platform/license
BitLocker rotation fail	Verify escrow state

Verification evidence

- Action status: Pending/succeeded/failed
- Audit actor: Who initiated
- Query result: Named columns and timestamp
- Diagnostics bundle: Collected logs
- Practice bridge: Lab 9.

3.9 Diagnose policy conflicts from device evidence

Mechanism

1. Assignment: Multiple profiles
2. CSP processing: Local MDM client
3. Per-setting merge: Winner or conflict
4. Reporting: Service aggregation
5. Remediation: Remove overlap

Exact contract or configuration

Field/state	Required interpretation
Client report	MDMDiagReport.html
Policy view	Per-setting status
Registry path	PolicyManager current values
Event log	Enterprise-Diagnostics-Provider

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Conflict	Two profiles set different values
Not applicable	Platform/edition mismatch
Error	CSP rejected value
Pending	No recent device check-in

Verification evidence

- Setting name: Exact CSP
- Source profiles: Both assignments
- Client event: Error code
- Retest: Success after consolidation
- Practice bridge: Lab 5.

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4. Protect devices (15–20%)

Alignment: LO2 · LO3 · K3 · K4 · A4 · A5 · A6. The following sections mirror the deck's mechanism → contract → failure/control → verification pattern.

4.1 Configure Defender Antivirus policy

Mechanism

1. Intune policy: Antivirus settings
2. CSP delivery: Defender configuration
3. Engine: Real-time protection
4. Cloud service: Block-at-first-sight
5. Telemetry: Protection status

Exact contract or configuration

Field/state	Required interpretation
Cloud protection	Enabled
Sample submission	Safe samples automatically
PUA	Block
Tamper protection	On through security management

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Setting locked	Tamper protection
Signature stale	Trigger intelligence update
Policy conflict	Security policy vs profile
Sensor unhealthy	Collect Defender diagnostics

Verification evidence

- Get-MpComputerStatus: Engine + signature
- Policy report: Applied settings
- Device timeline: Detection evidence
- Security intelligence: Current version
- Practice bridge: Lab 6.

4.2 Manage BitLocker encryption and recovery

Mechanism

1. Encryption policy: Algorithm and scope
2. TPM protector: Key material

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3. Volume encryption: Background state

4. Key escrow: Entra recovery object

5. Rotation: Invalidate exposed key

Exact contract or configuration

Field/state	Required interpretation
Silent enablement	Standard user allowed
Recovery escrow	Before encryption required
Algorithm	XTS-AES
Rotation	After key use

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
No escrow	Check device join and permissions
TPM not ready	Inspect tpm.msc
Encryption suspended	Resume protectors
Recovery key exposed	Rotate from Intune

Verification evidence

- manage-bde -status: Protection state
- Recovery object: Key ID match
- Encryption report: Compliant volume
- Rotation audit: New key timestamp
- Practice bridge: Lab 6.

4.3 Enforce Windows Firewall policy

Mechanism

1. Profile detection: Domain/private/public

2. Policy merge: Local + MDM rules

3. Filtering platform: Inbound/outbound decision

4. Logging: Dropped and allowed traffic

5. Report: Firewall state

Exact contract or configuration

Field/state	Required interpretation
Default inbound	Block
Rule scope	Program · port · address
Local merge	Disable where required
Log path	pfirewall.log

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Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Wrong profile	Inspect active network category
Overbroad rule	Constrain program and remote address
Rule conflict	Review policy stores
No log	Enable successful/dropped logging

Verification evidence

- Get-NetFirewallProfile: Enabled state
- Get-NetFirewallRule: Policy source
- Firewall log: Dropped packet
- Intune report: Profile applied
- Practice bridge: Lab 6.

4.4 Reduce attack surface with ASR and Zero Trust

Mechanism

1. Process behavior: Command or content
2. ASR rule engine: Rule ID evaluation
3. Mode: Audit · warn · block
4. Defender telemetry: Event recorded
5. Operations: Tune exclusions

Exact contract or configuration

Field/state	Required interpretation
Phased rollout	Audit then warn then block
Rule ID	Stable GUID
Exclusion	Narrow and justified
Zero Trust	Verify explicitly and limit impact

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Business app blocked	Use event evidence
Global exclusion	Replace with per-rule exclusion
No telemetry	Verify cloud protection
Rule not active	Check edition and policy precedence

Verification evidence

- Event 1121: Blocked action
- Advanced hunting: DeviceEvents

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- ASR report: Rule and device
- Exception register: Owner + expiry
- Practice bridge: Lab 6.

4.5 Apply security baselines without configuration drift

Mechanism

1. Baseline template: Recommended settings
2. Profile instance: Versioned configuration
3. Assignment: Pilot cohort
4. Per-setting report: State
5. Update workflow: Review new template

Exact contract or configuration

Field/state	Required interpretation
Baseline version	Immutable template snapshot
Pilot	Representative devices
Exception	Documented deviation
Migration	Create new profile then retire old

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Conflict	Same setting in another profile
Template update	Not automatic
Legacy setting	Unsupported on current OS
Overhardening	Validate business workload

Verification evidence

- Per-setting status: Success/conflict
- Baseline version: Recorded
- Pilot acceptance: Security + productivity
- Exception evidence: Approved control
- Practice bridge: Lab 6.

4.6 Integrate Intune with Defender for Endpoint

Mechanism

1. Connector: Tenant trust
2. Onboarding policy: Sensor activation
3. Risk signal: Defender device risk

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4. Compliance: Risk threshold

5. Conditional Access: Resource gate

Exact contract or configuration

Field/state	Required interpretation
Connector status	Active
EDR policy	Onboard blob
Risk level	Clear · low · medium · high
Compliance rule	Require below threshold

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Device not onboarded	Check Sense service
Duplicate record	Resolve device identity
Risk not flowing	Check connector scope
Access unexpectedly blocked	Inspect CA and risk timeline

Verification evidence

- MDE device page: Sensor health
- Intune connector: Last sync
- Compliance detail: Machine risk
- Sign-in result: Risk-based block
- Practice bridge: Lab 6.

4.7 Control application execution with App Control

Mechanism

1. Policy creation: Allow rules

2. Code integrity: Signature evaluation

3. Execution request: File identity

4. Decision: Allow/audit/block

5. Telemetry: CodeIntegrity events

Exact contract or configuration

Field/state	Required interpretation
Base policy	Trusted estate
Supplemental policy	Scoped extension
Audit mode	Inventory before enforcement
Managed installer	Trust tagged deployment source

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Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Unsigned business app	Create publisher/file rule
Policy brick risk	Stage in audit
Rule too broad	Avoid path trust where writable
Rollback unavailable	Maintain recovery plan

Verification evidence

- CodeIntegrity event: File and signer
- Policy ID: Applied CIPolicy
- Audit inventory: Required exceptions
- Block test: Unauthorized binary denied
- Practice bridge: Lab 6.

4.8 Operate update rings, Autopatch and Hotpatch

Mechanism

1. Release service: Quality/feature update
2. Deployment service: Target ring
3. Client scan: Applicability
4. Install/restart: Deadline behavior
5. Reporting: Update state

Exact contract or configuration

Field/state	Required interpretation
Ring	Deferral + deadline
Feature policy	Pinned version
Autopatch	Service-managed rings
Hotpatch	Eligible baseline + quarterly refresh

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Offer not received	Check safeguards and policy
Restart disruption	Tune active hours/deadlines
Conflicting policies	Consolidate update source
Hotpatch ineligible	Validate edition and baseline

Verification evidence

- Update report: Offer/install state
- Client log: WindowsUpdate.log

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- Autopatch status: Release phase
- KPI: Compliance by deadline
- Practice bridge: Lab 7.

5. Manage and secure applications (15–20%)

Alignment: LO2 · LO3 · K2 · K3 · A2 · A5. The following sections mirror the deck's mechanism → contract → failure/control → verification pattern.

5.1 Package and deploy Win32 applications

Mechanism

1. Source files: Install payload
2. Content prep tool: .intunewin
3. Service upload: Encrypted package
4. IME download: Device cache
5. Installer: Exit code + detection

Exact contract or configuration

Field/state	Required interpretation
Install command	Silent and idempotent
Detection	MSI · file · registry · script
Requirements	Architecture and OS
Return codes	Success · retry · reboot

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
0x87D30067	Detection failed after install
0x80070002	File or path missing
Installer context	System vs user mismatch
Reboot loop	Map return codes correctly

Verification evidence

- IME log: IntuneManagementExtension.log
- Detection output: Installed state
- App report: Device install status
- Exit code: Mapped result
- Practice bridge: Lab 8.

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5.2 Sequence apps with dependencies and supersedence

Mechanism

1. Assignment: Required/available/uninstall
2. Dependency graph: Prerequisites
3. Install order: Topological sequence
4. Supersedence: Update or replace
5. Detection: Final state

Exact contract or configuration

Field/state	Required interpretation
Dependency	Auto-install prerequisite
Supersedence	Update or uninstall previous
Requirement	Gate by device state
Availability	Company Portal intent

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Cycle	Remove circular dependency
Wrong uninstall	Test supersedence option
Detection overlap	Use version-aware rule
Assignment conflict	Resolve required vs uninstall

Verification evidence

- Dependency graph: Expected sequence
- IME log: Install orchestration
- App version: New build detected
- Report: No stale required failure
- Practice bridge: Lab 8.

5.3 Deploy Microsoft 365 Apps

Mechanism

1. Configuration: Apps and channel
2. Content source: Office CDN
3. Click-to-Run: Streaming install
4. Policy service: Cloud policy
5. Inventory: Version and channel

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Exact contract or configuration

Field/state	Required interpretation
Update channel	Current Enterprise or selected channel
Architecture	64-bit by default
ODT XML	Product · language · updates
Policy priority	Cloud policy vs local

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Channel mismatch	Review CDNBaseUrl
Office apps open	Handle update deadline
Network optimization	Use Delivery Optimization
Policy conflict	Inspect cloud policy status

Verification evidence

- OfficeC2RClient: Version/channel
- Intune app status: Installed
- Apps admin center: Inventory
- Policy report: Applied Office settings
- Practice bridge: Lab 8.

5.4 Integrate Apple and Managed Google Play stores

Mechanism

1. Store tenant link: ABM or Managed Google Play
2. App acquisition: Token/sync
3. Assignment: User/device/license
4. Managed install: Platform service
5. Inventory: Install status

Exact contract or configuration

Field/state	Required interpretation
Apple VPP token	Location-specific and renewable
Managed Google Play	Enterprise bind
App type	Managed store app
Licensing	Device or user where supported

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Token expired	Renew with same account

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App not synced	Force connector sync
License exhausted	Acquire more licenses
Wrong storefront	Verify tenant binding

Verification evidence

- Connector health: Last successful sync
- App record: Current metadata
- License count: Assigned/available
- Device report: Managed install
- Practice bridge: Lab 8.

5.5 Protect corporate data with MAM

Mechanism

1. App launch: User signs in
2. SDK/wrapper: Policy acquisition
3. Data boundary: Work identity context
4. Action evaluation: Save/copy/open-in
5. Selective wipe: Corporate data removed

Exact contract or configuration

Field/state	Required interpretation
Data transfer	Policy-managed apps only
Encryption	Organizational data
Access requirement	PIN + threat condition
Wipe	User or device scoped

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
App not protected	Unsupported app or wrong identity
Policy not applied	Check user assignment and license
Copy blocked unexpectedly	Review exempt apps
Wipe pending	App must check in

Verification evidence

- App protection status: User + app + device
- Policy diagnostics: Applied settings
- Selective wipe report: Completed
- CA result: Require app protection
- Practice bridge: Lab 8.

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5.6 Combine app protection with Conditional Access

Mechanism

1. Sign-in: User + app
2. CA conditions: Platform/client/risk
3. Grant control: Require approved app or APP
4. App policy: Data controls
5. Token: Protected session

Exact contract or configuration

Field/state	Required interpretation
Grant	Require app protection policy
Client apps	Mobile apps and desktop clients
Report-only	Validate before enforce
Exclusion	Emergency access accounts

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Unsupported client	Block or constrain
Policy gap	CA grants access without APP
Lockout	Use report-only and exclusions
Legacy auth	Block separately

Verification evidence

- Sign-in log: CA policy result
- APP report: Policy delivered
- Token claim: Client/app context
- Test matrix: Managed/unmanaged × platform
- Practice bridge: Lab 8.

5.7 Deliver app configuration to managed and unmanaged apps

Mechanism

1. Configuration source: Intune key-value payload
2. Assignment: User or device
3. Delivery channel: MDM or MAM
4. App reads keys: Runtime behavior
5. Reporting: Configuration status

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Exact contract or configuration

Field/state	Required interpretation
Managed device	Configuration through MDM
Managed app	Configuration through APP channel
Data type	String · boolean · integer
Vendor schema	Exact supported keys

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Key ignored	Wrong name or type
Wrong channel	Managed app vs managed device
Conflicting values	Consolidate policies
No report	Verify app supports reporting

Verification evidence

- Policy payload: Exact keys
- App diagnostics: Received configuration
- Device report: Success
- Behavior test: Expected tenant/account state
- Practice bridge: Lab 8.

5.8 Troubleshoot application deployment from logs

Mechanism

1. Assignment resolution: Intent
2. IME policy sync: App metadata
3. Content download: DO/BITS
4. Installer execution: Context
5. Detection: Reported state

Exact contract or configuration

Field/state	Required interpretation
IME log	IntuneManagementExtension.log
Agent Executor	PowerShell execution
Content cache	Staging files
Portal report	Device install details

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Download timeout	Network/proxy/DO

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Install failure	Exit code and command
Detection false negative	Fix rule
Requirement not met	Inspect applicability

Verification evidence

- Timestamp chain: First failing phase
- Exit code: Vendor meaning
- Detection trace: Rule output
- Retest: Successful device status
- Practice bridge: Lab 8.

5.9 Manage app lifecycle and inventory

Mechanism

1. Discovery: Managed app inventory
2. Classification: Owner + criticality
3. Deployment: Install intent
4. Update: Supersedence/store update
5. Retirement: Uninstall and revoke

Exact contract or configuration

Field/state	Required interpretation
Owner	Named service owner
Detection	Version-aware
Update SLA	Risk-based deadline
Retirement	Data and access cleanup

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Shadow IT	Compare discovered vs approved
Stale package	Monitor version drift
Uninstall fails	Validate command/context
Orphan assignment	Review group lifecycle

Verification evidence

- Discovered apps: Version footprint
- Install base: Device/user count
- Update compliance: Current version %
- Retirement proof: No active assignment
- Practice bridge: Lab 8.

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6. Optimize endpoint operations (10–15%)

Alignment: LO3 · LO4 · K3 · K5 · A5 · A7 · A8. The following sections mirror the deck's mechanism → contract → failure/control → verification pattern.

6.1 Automate Intune with Microsoft Graph

Mechanism

1. Identity: App or delegated sign-in
2. OAuth token: Scope/role claims
3. Graph endpoint: Resource request
4. Pagination: @odata.nextLink
5. Result handling: Retry and audit

Exact contract or configuration

Field/state	Required interpretation
Base URI	https://graph.microsoft.com/v1.0
Permission	DeviceManagementManagedDevices.Read.All
Pagination	Follow every nextLink
Throttling	Honor Retry-After

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
401	Token missing/expired
403	Permission or role absent
429	Back off and retry
Partial export	Pagination ignored

Verification evidence

- Token claims: aud + roles/scp
- HTTP trace: Status + request-id
- Count check: All pages retrieved
- Audit log: Automation identity
- Practice bridge: Lab 9.

6.2 Use PowerShell for repeatable endpoint operations

Mechanism

1. Connect-MgGraph: Authentication
2. Select-MgProfile: API surface

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3. Get cmdlet: Read state

4. Update cmdlet: Controlled mutation

5. Export/log: Evidence and rollback

Exact contract or configuration

Field/state	Required interpretation
Context	Get-MgContext
Least privilege	Explicit scopes
WhatIf pattern	Preview intended targets
Idempotence	Compare before update

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Wrong tenant	Check TenantId
Broad filter	Count before mutate
Module drift	Pin/test version
Transient failure	Retry safe operations

Verification evidence

- Transcript: Command and time
- Before/after JSON: Changed fields
- Target count: Expected devices
- Exit status: Automation result
- Practice bridge: Lab 9.

6.3 Extend compliance with custom scripts

Mechanism

1. Discovery script: Return JSON

2. Custom schema: Setting definitions

3. Compliance engine: Compare values

4. Device status: Per-setting result

5. Remediation: Operational response

Exact contract or configuration

Field/state	Required interpretation
Exit code	0 for successful discovery
Output	Single compressed JSON object
Data types	Match schema
Schedule	Periodic reevaluation

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Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Invalid JSON	Validate serialization
Wrong property type	Match boolean/int/string
Script timeout	Keep discovery bounded
Sensitive data	Do not emit secrets

Verification evidence

- Local script output: Valid JSON
- Per-setting report: Detected value
- Compliance state: Expected result
- Remediation ticket: Closed with proof
- Practice bridge: Lab 9.

6.4 Operate Security Copilot agents responsibly

Mechanism

1. Signal intake: Threat/performance data
2. Agent analysis: Pattern and context
3. Recommendation: Suggested action
4. Human review: Validate scope and evidence
5. Execution: Controlled remediation

Exact contract or configuration

Field/state	Required interpretation
Grounding	Tenant telemetry
Recommendation	Not automatic authority
Review	Check affected devices and blast radius
Audit	Record decision and outcome

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Hallucinated cause	Verify against raw evidence
Overbroad action	Pilot and approve
Sensitive prompt	Apply data governance
Agent unavailable	Maintain manual runbook

Verification evidence

- Source evidence: Linked telemetry
- Reviewer decision: Accept/modify/reject

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- Change record: Approved scope
- Outcome metric: Issue resolved without regression
- Practice bridge: Lab 10.

6.5 Build operational reports and dashboards

Mechanism

1. Service data: Device/app/policy facts
2. Report query: Filters and columns
3. Export/API: Structured data
4. Workbook/dashboard: Operational view
5. Decision: Action and owner

Exact contract or configuration

Field/state	Required interpretation
Freshness	Record report timestamp
Filter	Define cohort explicitly
Export	CSV or Graph
KPI	Numerator, denominator, threshold

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Misleading denominator	Exclude not-applicable correctly
Stale export	Display freshness
Duplicate devices	Use stable device ID
Vanity metric	Tie KPI to action

Verification evidence

- Query definition: Reproducible filter
- Dashboard tile: Current value
- Drill-through: Affected devices
- Action log: Owner and deadline
- Practice bridge: Lab 9.

6.6 Interpret Endpoint Analytics and reliability

Mechanism

1. Device telemetry: Startup/app/restart events
2. Aggregation: Population metrics
3. Scoring: Startup + application reliability

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4. Insight: Outlier or regression

5. Action: Hardware/policy/remediation

Exact contract or configuration

Field/state	Required interpretation
Startup score	Boot + sign-in experience
App reliability	Crashes and hangs
Restart frequency	Operational stability
Baseline	Compare cohort and time

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Low sample	Avoid overinterpreting
Cohort mix	Segment by model/OS
Score improved but tickets rise	Use multiple signals
Telemetry gap	Check prerequisites

Verification evidence

- Score trend: Before/after
- Model cohort: Outlier devices
- App report: Top regressions
- Validation: Reduced incident rate
- Practice bridge: Lab 9.

6.7 Deploy proactive remediations safely

Mechanism

1. Detection script: Exit 0/1

2. Service evaluation: Issue found

3. Remediation script: Correct state

4. Post-detection: Verify fix

5. Reporting: Run outcome

Exact contract or configuration

Field/state	Required interpretation
Detection	No change, evidence only
Remediation	Idempotent and bounded
Execution	System or user context
Schedule	Operational need

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Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Detection always fails	Fix exit logic
Remediation loops	Make desired state idempotent
32/64-bit mismatch	Choose host setting
Large blast radius	Pilot and stage

Verification evidence

- Pre-state: Issue reproduced
- Detection result: Issue found
- Remediation log: Change made
- Post-state: Issue absent
- Practice bridge: Lab 9.

6.8 Monitor tenant health, alerts and change

Mechanism

1. Service signal: Health/message center
2. Operational baseline: Expected rate
3. Alert rule: Threshold or event
4. Triage: Impact and ownership
5. Response: Communicate and recover

Exact contract or configuration

Field/state	Required interpretation
Enrollment alert	Failure-rate threshold
Compliance drift	Noncompliant percentage
Config conflict	New conflict count
Service advisory	Affected workload and region

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Alert fatigue	Tune severity and routing
No baseline	Measure normal first
Duplicate incident	Correlate service advisory
Unowned action	Use escalation matrix

Verification evidence

- Alert payload: Cohort + timestamp
- Incident timeline: Detection to recovery

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- Change correlation: Policy/deployment event
- Postmortem: Control improvement
- Practice bridge: Lab 10.

6.9 Investigate unauthorized endpoint access

Mechanism

1. Detection: Risk/sign-in/device alert
2. Scope: User + device + app
3. Containment: Disable/revoke/isolate
4. Evidence: Logs and timeline
5. Recovery: Restore and monitor

Exact contract or configuration

Field/state	Required interpretation
Identity action	Revoke sessions
Device action	Isolate or retire
Evidence handling	Preserve timestamps and IDs
Access review	Remove excess privilege

Failure, diagnosis and recovery

Failure signal	Diagnostic or recovery rule
Premature wipe	Preserve evidence first
Incomplete scope	Correlate device and sign-in IDs
Credential persists	Reset and revoke tokens
Access restored early	Require clean compliance

Verification evidence

- Incident timeline: Ordered events
- Sign-in log: CA and device details
- Device timeline: Alert/process activity
- Closure proof: Risk cleared and access reviewed
- Practice bridge: Lab 10.

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7. Detailed hands-on labs

Lab 1: Entra device identity, dynamic groups and delegated administration

Duration: 110 minutes · Alignment: LO1 · K1 · K4 · A1 · A3

Deliverable: Validated join-state evidence pack, dynamic-group rule, scoped role assignment and audit trace

Safety and prerequisites

- Use only the training tenant, test users and test devices.
- Confirm the tenant ID before changing state.
- Capture before-state evidence and use least privilege.
- Ask before wipe, retire, isolate or reprovision actions.

Establish tenant and role boundaries

1. Sign in to the Microsoft Entra admin center with the lab administrator role supplied by the trainer.
2. Record the tenant ID and confirm you are in the training tenant, not a production tenant.
3. Open Identity > Devices > All devices and record the current device count.
4. Open Devices > Device settings and document join and local-administrator controls before changing them.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Validate a device identity

5. On the Windows test device, open Windows Terminal as administrator.
6. Run `dsregcmd /status` and save the Device State, Device Details and SSO State blocks.
7. Correlate DeviceId and TenantId with the Entra device object.
8. Classify the device as registered, Entra joined or hybrid joined and justify the classification from flags.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Create a dynamic targeting rule

9. Open Entra ID > Groups > New group and select Security with Dynamic Device membership.
10. Create a rule for corporate Windows devices, preview the result, then save only after the expected test device appears.
11. Record the exact rule and preview result in the evidence folder.
12. Assign a descriptive owner and purpose to the group.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Delegate and verify

13. Create or inspect an Intune role assignment with a narrow scope group and scope tag.

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14. Use a test operator account to verify visibility is restricted to intended objects.
15. Review the audit log for the role/group changes.
16. Revoke the temporary test assignment if instructed and capture the final state.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Evidence submission

Save evidence under labs/lab-01-entra-identity-groups-rbac/evidence/. Remove passwords, recovery keys, access tokens and unrelated personal data.

Lab 2: Multi-platform Intune enrollment and troubleshooting

Duration: 115 minutes · Alignment: LO1 · K1 · K3 · A1 · A3

Deliverable: Enrollment configuration matrix, MDMDiag evidence, error-code diagnosis and successful retest

Safety and prerequisites

- Use only the training tenant, test users and test devices.
- Confirm the tenant ID before changing state.
- Capture before-state evidence and use least privilege.
- Ask before wipe, retire, isolate or reprovision actions.

Plan enrollment

17. Build a platform matrix for Windows, macOS, iOS/iPadOS and Android corporate/personal ownership.
18. Record the intended enrollment method, authority connector, license and user scope for each platform.
19. In Intune, review Devices > Enrollment and record token/connector health.
20. Review enrollment restrictions and identify which personal platforms are allowed or blocked.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Configure the Windows path

21. Verify MDM user scope includes the lab user.
22. Verify the user has the required Intune entitlement.
23. Join or register the Windows test device according to the lab scenario.
24. Trigger Settings > Accounts > Access work or school > Info > Sync.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Collect diagnostics

25. Run `mdmdiagnosticstool.exe -area DeviceEnrollment;DeviceProvisioning;Autopilot -cab evidence\MDMDiagnostics.cab`.
26. Open Event Viewer > DeviceManagement-Enterprise-Diagnostics-Provider > Admin.

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27. Record the first error code or successful enrollment event and its timestamp.
28. Use Intune Troubleshooting + support to correlate the user and managed-device record.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Seed and recover

29. Apply the trainer-provided restriction or scope mismatch that blocks enrollment.
30. Reproduce the failure and record the exact HRESULT/correlation ID.
31. Correct only the failed prerequisite, then retry enrollment.
32. Capture the successful managed record, ownership, primary user and last check-in.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Evidence submission

Save evidence under labs/lab-02-intune-enrollment-troubleshooting/evidence/. Remove passwords, recovery keys, access tokens and unrelated personal data.

Lab 3: Compliance, Conditional Access, Windows Hello and LAPS

Duration: 115 minutes · Alignment: LO2 · K3 · K4 · A4 · A6

Deliverable: Compliance/CA decision trace, WHfB state and auditable LAPS recovery/rotation evidence

Safety and prerequisites

- Use only the training tenant, test users and test devices.
- Confirm the tenant ID before changing state.
- Capture before-state evidence and use least privilege.
- Ask before wipe, retire, isolate or reprovision actions.

Create compliance policy

33. In Intune, create a Windows compliance policy requiring encryption, firewall and the trainer-specified OS version.
34. Set a short training grace period and assign only to the lab device group.
35. Record every configured setting and the noncompliance action.
36. Sync the device and inspect per-setting compliance.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Create Conditional Access safely

37. In Entra Conditional Access, create a policy requiring a compliant device for the lab cloud app.
38. Exclude the designated emergency-access account.
39. Start in report-only mode and run the What If tool for compliant and noncompliant scenarios.
40. Enable only when the trainer confirms the report-only results are correct.

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Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Validate WHfB and LAPS

41. Assign the Windows Hello for Business policy and record the selected trust model.
42. On the device, run `dsregcmd /status` and verify the expected NGC/PRT state.
43. Assign the Windows LAPS policy to back up passwords to Entra ID.
44. Verify the password backup timestamp and retrieve it only with the authorized role.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Diagnose and prove recovery

45. Seed one compliance or access mismatch.
46. Use the device compliance details and sign-in log to locate the failed condition.
47. Correct the policy/assignment, sync, and rerun the sign-in test.
48. Rotate the LAPS password after access and capture the audit event.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Evidence submission

Save evidence under labs/lab-03-compliance-ca-whfb-laps/evidence/. Remove passwords, recovery keys, access tokens and unrelated personal data.

Lab 4: Autopilot, device preparation and Windows 365 provisioning

Duration: 115 minutes · Alignment: LO1 · LO4 · K1 · K5 · A2 · A7

Deliverable: Deployment decision, profile/ESP design, readiness evidence and Cloud PC provisioning check

Safety and prerequisites

- Use only the training tenant, test users and test devices.
- Confirm the tenant ID before changing state.
- Capture before-state evidence and use least privilege.
- Ask before wipe, retire, isolate or reprovision actions.

Choose deployment architecture

49. Compare classic Autopilot and Windows Autopilot device preparation against the lab constraints.
50. Choose user-driven, pre-provisioning or self-deploying mode and document prerequisites.
51. Define the Entra join type and device ownership state.
52. Create a rollback path that does not depend on user access.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

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Configure identity and profile

53. Create/import the lab device identity or configure the device preparation policy.
54. Assign the intended deployment profile to the test cohort.
55. Configure a valid device-name template.
56. Confirm the service shows the profile as assigned before starting OOBE.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Control provisioning

57. Create an Enrollment Status Page that blocks only the required security and application payload.
58. Record timeout and failure behavior.
59. Run the deployment on the approved test device or use the training simulation data.
60. Capture phase duration, blocking app status and final join/management state.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Extend to Windows 365

61. Create or inspect a provisioning policy with the selected image, join type and network.
62. Validate the Azure network connection or Microsoft-hosted network choice.
63. Assign the policy to the training user group.
64. Capture the Cloud PC provisioning state and Intune managed-device record.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Evidence submission

Save evidence under labs/lab-04-autopilot-device-prep-windows365/evidence/. Remove passwords, recovery keys, access tokens and unrelated personal data.

Lab 5: Settings Catalog, ADMX migration and assignment filters

Duration: 110 minutes · Alignment: LO2 · K2 · K4 · A2 · A4

Deliverable: Migrated settings policy, filter preview, conflict diagnosis and corrected per-setting result

Safety and prerequisites

- Use only the training tenant, test users and test devices.
- Confirm the tenant ID before changing state.
- Capture before-state evidence and use least privilege.
- Ask before wipe, retire, isolate or reprovision actions.

Analyze legacy policy

65. Export the trainer-provided GPO report as XML.
66. Import it into Group Policy analytics and record supported, deprecated and unknown settings.
67. Select only supported controls required by the scenario.
68. Record redesign decisions for unsupported settings.

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Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Create cloud policy

69. Build a Settings Catalog profile with the selected controls.
70. Import the required ADMX only if the setting is absent from the catalog.
71. Assign the policy to a pilot group.
72. Create an assignment filter using an exact supported device property.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Seed a conflict

73. Assign a second training profile that sets one control differently.
74. Sync the device and reproduce the per-setting conflict.
75. Collect the policy report, MDMDiag output and relevant MDM event.
76. Identify both source profiles before changing either.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Consolidate and verify

77. Choose the authoritative profile and remove the overlapping setting from the other profile.
78. Sync the device and wait for the service report to refresh.
79. Verify the per-setting state is successful and the client value matches.
80. Save the final policy export and filter preview as evidence.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Evidence submission

Save evidence under labs/lab-05-configuration-admx-filters/evidence/. Remove passwords, recovery keys, access tokens and unrelated personal data.

Lab 6: Endpoint security, Defender, BitLocker, firewall and ASR

Duration: 120 minutes · Alignment: LO2 · LO3 · K3 · K4 · A4 · A5 · A6

Deliverable: Security control set with Defender/BitLocker/firewall/ASR verification and exception record

Safety and prerequisites

- Use only the training tenant, test users and test devices.
- Confirm the tenant ID before changing state.
- Capture before-state evidence and use least privilege.
- Ask before wipe, retire, isolate or reprovision actions.

Design the control set

81. Define Defender Antivirus, BitLocker, Firewall and ASR requirements for the lab device group.

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82. Map each requirement to an Intune Endpoint security policy surface.
83. Identify where a security baseline overlaps and remove duplicate ownership.
84. Record the rollout mode for every ASR rule: audit, warn or block.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Deploy and inspect

85. Create and assign the approved endpoint-security policies to the pilot cohort.
86. Sync the test device.
87. Run `Get-MpComputerStatus`, `manage-bde -status` and `Get-NetFirewallProfile`.
88. Capture Intune per-setting status and Defender onboarding/sensor health.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Generate evidence

89. Run the trainer-approved benign ASR test or inspect provided event data.
90. Open the CodeIntegrity/Defender/Firewall evidence channels specified by the trainer.
91. Correlate device ID, policy name, event ID and timestamp.
92. Record the control that produced each observed outcome.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Tune safely

93. Seed an allowed business-app exception scenario.
94. Replace any broad global exclusion with the narrowest supported per-rule exception.
95. Retest the application and the control.
96. Create an exception record with owner, rationale, expiry and compensating control.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Evidence submission

Save evidence under labs/lab-06-endpoint-security-defender/evidence/. Remove passwords, recovery keys, access tokens and unrelated personal data.

Lab 7: Update rings, feature updates, Autopatch, Hotpatch and Delivery Optimization

Duration: 110 minutes · Alignment: LO1 · LO2 · K1 · K2 · A2 · A5

Deliverable: Staged update design, eligibility checks, rollout metrics and rollback decision evidence

Safety and prerequisites

- Use only the training tenant, test users and test devices.
- Confirm the tenant ID before changing state.
- Capture before-state evidence and use least privilege.
- Ask before wipe, retire, isolate or reprovision actions.

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Design staged rollout

97. Create pilot, early adopter and broad deployment cohorts.
98. Define quality-update deferral, deadline, grace period and restart experience per ring.
99. Create a feature update policy for the target Windows release.
100. Record rollback and pause decision thresholds.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Evaluate service options

101. Compare self-managed update rings with Windows Autopatch for the scenario.
102. Check Hotpatch edition, baseline and service prerequisites.
103. Configure Delivery Optimization group or connected-cache behavior for the lab network.
104. Document non-Windows update controls for macOS/iOS or Android where relevant.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Analyze update evidence

105. Open the Intune Windows update report and filter to the lab cohort.
106. Record offer, download, install, restart and error states.
107. On the device, generate WindowsUpdate.log or inspect the provided log sample.
108. Identify any safeguard hold, eligibility block or policy conflict.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Decide and verify

109. Apply the trainer-provided update failure scenario.
110. Choose pause, expedite, rollback or targeted remediation based on evidence.
111. Implement the approved action in the training tenant.
112. Capture the post-action compliance rate and affected-device list.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Evidence submission

Save evidence under labs/lab-07-updates-autopatch-hotpatch/evidence/. Remove passwords, recovery keys, access tokens and unrelated personal data.

Lab 8: Win32 apps, Microsoft 365 Apps, MAM and app configuration

Duration: 120 minutes · Alignment: LO2 · LO3 · K2 · K3 · A2 · A5

Deliverable: Deployable app contract, detection evidence, MAM/CA test matrix and troubleshooting trace

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Safety and prerequisites

- Use only the training tenant, test users and test devices.
- Confirm the tenant ID before changing state.
- Capture before-state evidence and use least privilege.
- Ask before wipe, retire, isolate or reprovision actions.

Build a Win32 app contract

113. Package the trainer-provided source with the Intune Win32 Content Prep Tool or inspect the supplied `.intunewin` metadata.
114. Define silent install and uninstall commands with the correct context.
115. Create version-aware detection and explicit requirement rules.
116. Map vendor exit codes to success, retry, soft reboot and hard reboot.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Deploy and sequence

117. Upload/create the app in Intune and assign it to the pilot group.
118. Configure the supplied prerequisite as a dependency.
119. Configure supersedence for the previous version without creating a cycle.
120. Trigger device sync and monitor install status.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Protect app data

121. Create an app protection policy for the supported mobile app.
122. Set work-data transfer, encryption, access and selective-wipe controls.
123. Create a Conditional Access policy requiring app protection in report-only mode.
124. Test managed and unmanaged device/app combinations using the evidence matrix.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Troubleshoot and close

125. Seed a detection or requirement-rule failure.
126. Inspect IntuneManagementExtension.log and the per-device install details.
127. Correct the smallest failing contract element and retry.
128. Capture installed version, detection success and app-protection status.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Evidence submission

Save evidence under labs/lab-08-apps-mam-configuration/evidence/. Remove passwords, recovery keys, access tokens and unrelated personal data.

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Lab 9: Graph, PowerShell, remote actions and Endpoint Analytics

Duration: 115 minutes · Alignment: LO3 · LO4 · K3 · K5 · A5 · A7

Deliverable: Least-privilege automation transcript, paginated export, remote-action evidence and KPI readout

Safety and prerequisites

- Use only the training tenant, test users and test devices.
- Confirm the tenant ID before changing state.
- Capture before-state evidence and use least privilege.
- Ask before wipe, retire, isolate or reprovision actions.

Authenticate with least privilege

129. Open PowerShell 7 and connect with `Connect-MgGraph` using only the lab read scopes.
130. Run `Get-MgContext` and verify TenantId, Account and Scopes.
131. Record the Graph `v1.0` endpoint and selected resource.
132. Do not request write permissions until the trainer authorizes the change task.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Export all managed devices

133. Request the managedDevices collection with selected fields.
134. Follow every `@odata.nextLink` until no next page remains.
135. Export device ID, name, OS, compliance and last-sync values to CSV.
136. Compare the exported count with the service report.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Perform a controlled action

137. Resolve one lab device by stable device ID.
138. Preview the intended action and confirm ownership.
139. Run the approved sync or rotate-key action.
140. Capture the audit actor, time and resulting status.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Interpret analytics

141. Open Endpoint Analytics and record the current startup, application reliability and restart signals.
142. Segment by the trainer-provided device model or OS cohort.
143. Identify one outlier and propose a bounded proactive remediation.
144. Define the before/after metric that will prove improvement.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

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Evidence submission

Save evidence under labs/lab-09-graph-powershell-analytics/evidence/. Remove passwords, recovery keys, access tokens and unrelated personal data.

Lab 10: Capstone: investigate access, remediate endpoints and brief operations

Duration: 110 minutes · Alignment: LO4 · K5 · A7 · A8

Deliverable: Incident timeline, containment record, remediation proof, access review and executive operations brief

Safety and prerequisites

- Use only the training tenant, test users and test devices.
- Confirm the tenant ID before changing state.
- Capture before-state evidence and use least privilege.
- Ask before wipe, retire, isolate or reprovision actions.

Triage unauthorized access

145. Review the supplied sign-in, Conditional Access, device risk and endpoint timeline evidence.
146. Correlate user ID, device ID, app, IP, policy and timestamps.
147. Build an ordered incident timeline.
148. State the scope and confidence level before containment.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Contain and preserve evidence

149. Revoke user sessions and isolate/retire only the device authorized by the scenario.
150. Preserve relevant logs, request/correlation IDs and screenshots before destructive actions.
151. Remove or suspend excessive access rights.
152. Record every containment action, actor and time.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Remediate operations

153. Use the supplied detection script to confirm the endpoint issue.
154. Run the approved idempotent remediation on the pilot device.
155. Perform a post-detection or direct state check.
156. Restore access only after identity risk, device health and compliance are acceptable.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Brief and close

157. Create a concise operations brief: impact, root cause, control failure, actions and remaining risk.
158. Attach the evidence index and access-review result.

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159. Define a tenant-health alert or operational baseline that would detect recurrence sooner.
160. Obtain trainer sign-off on A7/A8 evidence.

Acceptance check: Capture the configured state, observed result, timestamp/correlation ID and interpretation.

Evidence submission

Save evidence under labs/lab-10-capstone-incident-operations/evidence/. Remove passwords, recovery keys, access tokens and unrelated personal data.

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8. Assessment preparation

- Written Assessment: five open-ended questions covering K1–K5 in 60 minutes.
- Practical Performance: six connected tasks covering A1–A8 in 60 minutes.
- Practice with the same evidence chain used in labs: state, result, diagnosis and recovery.
- Use the official Microsoft study guide as the authority for current exam scope.

9. Sources and currency

Certification: https://learn.microsoft.com/en-us/credentials/certifications/modern-desktop/

Study Guide: https://learn.microsoft.com/en-us/credentials/certifications/resources/study-guides/md-102

Course: https://learn.microsoft.com/en-us/training/courses/md-102t00

Labs: https://github.com/MicrosoftLearning/MD-102T00-Microsoft-365-Endpoint-Administrator

Lab Commit: 25eaad2a79548c4767ea45eae8d08649622c583a

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